EE3355: SIGNALS AND SYSTEMS

LABORATORY 03 – PRE-LAB

NAME : AKARSHANA A.N.
REG. NO : EG/2021/4392
GROUP NO : N/A
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1. PRE-LAB QUESTIONS

1.1 Explain about the analog to digital conversion.

Analog to digital conversion (ADC) turns a continuous-time, continuous-amplitude signal into a sequence of binary words that a digital system can store and process. It consists of three steps:

1. **Sampling:** a sample-and-hold unit captures the signal value at regular intervals and holds it constant for one sampling period. To avoid aliasing, the sampling frequency must be at least twice the highest frequency in the signal (Nyquist criterion).
2. **Quantization:** each held value is mapped to the nearest of 2^N discrete levels, where N is the number of bits. This introduces a quantization error of up to half of one step.
3. **Encoding:** each quantization level is represented by an N-bit binary word (for example, 0000 to 1111 for four bits).

The resolution of an N-bit converter is given by $\text{Res\%} = V_{ref} / (2^N - 1) \times 100\%$, so increasing the number of bits makes each step smaller and the digital output a closer representation of the input. Common ADC architectures include parallel (flash), successive approximation, dual-slope and sigma-delta converters. The parallel comparator converter used in this laboratory compares the input against several reference levels at the same time using op-amp comparators, and an encoding unit converts the comparator outputs into a binary word. It is the fastest architecture, but it needs $2^N - 1$ comparators.

1.2 What is the function of PWM in D/A conversion?

Pulse Width Modulation (PWM) allows a digital output that can only be fully high or fully low to produce an analog voltage. The digital value to be converted sets the duty cycle of a fixed-frequency pulse train: a larger value gives a wider high pulse in each period.

A low-pass filter (usually a simple RC filter) then averages the pulse train. The filter output is a DC voltage proportional to the duty cycle:

$$V_{out} \approx D \times V_{high}, \quad D = t_{on} / T$$

Therefore, PWM replaces a resistor network or dedicated DAC chip with a single digital pin, a timer and a filter, which is why it is widely used in microcontrollers. Its resolution is set by the number of timer bits, while the PWM frequency must be much higher than the filter cut-off frequency so that the ripple on the output is small. The trade-off is a slower response, since the filter must settle after every change in duty cycle.

1.3 What are the applications of Digital to Analog Converters?

- **Audio systems:** music players, phones and sound cards convert stored digital audio to the analog signal that drives speakers and headphones.
- **Video and displays:** converting digital image data to analog signals for display drivers and analog video outputs.
- **Communication systems:** modems, software-defined radio and transmitters use DACs to generate the analog waveform that is sent through the channel.
- **Control systems and motor drives:** a digital controller outputs analog command signals to actuators, valves and motor drives.
- **Test and measurement:** arbitrary waveform generators and function generators produce analog test signals from stored digital samples.
- **Medical and industrial equipment:** ultrasound, patient monitors and process instrumentation use DACs for signal generation and calibration.